[INCH-POUND]
A-A-59392
15 December 1999
SUPERSEDING
MIL-G-80163B
August 22, 1986

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINES, SURFACE, ROTARY TABLE, HORIZONTAL SPINDLE

The General Services Administration has authorized the use of this commercial item description as a replacement for all federal agencies.

1. SCOPE. This commercial item description (CID) covers horizontal spindle rotary table surface grinding machines for use in batch grinding or tool and die applications where concave, convex, precision, and fine-finish grinding is required.

2. CLASSIFICATION. The horizontal spindle rotary table surface grinding machines shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 7.2(b)).

Type I - Vertical traversing wheelhead
Type II - Horizontal traversing wheelhead
Type III - Vertical and horizontal traversing wheelhead

Size: Sizes are listed in tables I through III.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

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TABLE I. Type I, machine sizes.

Size	Table diameter, inches
16	16
20	20
24	24
30	30
35	35
39	39
47	47
55	55
59	59
69	69

TABLE II. Type II, machine sizes.

Size	Table diameter, inches
16	16
20	20
24	24
30	30
44	44

TABLE III. Type III, machine sizes.

Size	Table diameter, inches
12	12
14	14
16	16
20	20
24	24
30	30
39	39
47	47
65	65

Note: The range of machine sizes shown in tables I, II, and III imply that the machine offered shall be equal to or larger than the size stated but smaller than the next available size. When the largest machine size is ordered, the machine offered shall not exceed the stated requirements by more than 10 percent.

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The grinding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. The grinding machine shall have a minimum workpiece height capacity as specified (see 7.2(c)). The machine shall be of sufficient strength and rigidity to withstand all forces encountered during operation of the grinding machine at maximum capacity and grinding speed while maintaining the alignment requirements specified in section 3.3.

3.2 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Base. A means shall be provided for leveling and anchoring the machine to the floor. A means shall be provided for collection and discharge of coolant and grinding swarf from the base. A means shall be provided for access to serviceable components housed within the base.

3.2.2 Wheelhead assembly. The wheelhead assembly shall include a wheelhead housing, spindle, spindle drive unit, positioning and feed mechanism, and position and feed drive unit. Unless otherwise specified (see 7.2(d)), the spindle drive shall provide peripheral grinding wheel speeds within the range of 3,400 to 6,000 surface feet per minute. The spindle shall be mounted on antifriction bearings designed to minimize axial and radial play and shall be dynamically balanced. The spindle bearings shall be continuously or permanently lubricated. Unless otherwise specified (see 7.2(e)), on type I and III machines the wheelhead shall vertically traverse 120 percent of the workpiece height capacity. On type II and III machines the wheelhead shall horizontally traverse a minimum of half the diameter of the rotary table.

3.2.3 Machineways. Machineways shall be protected regardless of grinding machine position to exclude chips, dust, coolant, and other contaminants or hazards that may cause damage or result in inaccurate performance.

3.2.4 Grinding wheel. Grinding wheels shall meet the requirements of American National Standards Institute (ANSI) B7.1, ANSI B74.1, ANSI B74.13 and ANSI B74.2. A grinding wheel(s) shall be provided as specified (see 7.2(f)).

3.2.5 Rotary table. The rotary table shall be a circular table oriented in the horizontal plane. Unless otherwise specified (see 7.2(g)), the minimum workpiece weight capacity of the rotary table shall meet the requirements of table IV. If required, the table shall be capable of horizontal tilt adjustment as specified (see 7.2(h)). The table on type I machines shall horizontally traverse a minimum of half the diameter of the table. The table on type II machines shall vertically traverse a minimum of 120 percent of the full workpiece height capacity. The table on type III machines shall not traverse. Unless otherwise specified (see 7.2(i)), table rotation shall be variable and have a peripheral speed of at least 280 surface feet per minute. Independent drive units and controls shall be provided for table rotation, and when applicable, for table transition.

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TABLE IV. Workpiece weight capacity of rotary table.

Type	Capacity, pounds per square foot of work area
I	75
II	100
III	150

3.2.6 Magnetic chuck. Unless otherwise specified (see 7.2(j)), the magnetic chuck shall be the same diameter as and be securely mounted on the rotary table. When mounted, the magnetic chuck shall be interlocked to prevent chuck/table rotation until magnetized. Unless otherwise specified (see 7.2(k)), the magnetic poles shall not be greater than 1/4 inch apart over the entire face of the chuck. The magnetic chuck shall securely hold the workpiece to the face of the chuck during maximum metal removal conditions using the manufacturer's recommended loading and blocking procedures.

3.2.7 Wheel dresser. Unless otherwise specified (see 7.2(l)), a wheel dresser capable of dressing the entire working face of the grinding wheel shall be provided.

3.2.8 Drive system. All drive system elements (pumps, electric motors, hydraulic power takeoff, or other means of energy-to-motion converters) shall be capable of continuously operating the grinding machine at maximum speed, grinding load, and capacity without exceeding 85 percent of the rated horsepower of any of the individual elements.

3.2.9 Controls. The grinding machine shall have manual (mechanical, electrical, or both) positioning and feed. If required, the grinding machine shall be capable of semiautomatic operation, fully automatic operation, or both.

3.2.9.1 Semiautomatic cycle. When specified (see 7.2(m)), the grinding machine shall be provided with semiautomatic cycle operation. After manually inputting predetermined machine and workpiece parameters, loading the workpiece, and activating the magnetic chuck, the operator shall complete a grind cycle by starting the cycle from a single control. The semiautomatic cycle shall include, as a minimum, the following functions:

a. Rapid advance to the grinding position.
b. Starting chuck/table rotation.
c. Starting grinding wheel rotation.
d. Engaging coolant pump.
e. Engaging power feed at preset rate until stopped by automatic gauging.
f. Rapid return to preset location.
g. Stopping the grinding wheel rotation and coolant pump.
h. Retracting the chuck/table to the load position (when applicable), stopping chuck/table rotation, and demagnetizing the chuck.

3.2.9.2 Fully automatic cycle. When specified (see 7.2(n)), the grinding machine shall be provided with fully automatic cycle operation. After starting the grinding machine, the only operator action required shall be to keep the hopper or other parts feed device supplied with parts. The machine shall automatically receive, position, grind to size, and eject the part. The automatic cycle shall also dress the wheel, compensate for wheel size reduction, and allow for variable sparkout.

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3.2.10 Dials and indicating devices. Unless otherwise specified (see 7.2(o)), all dials and indicating devices for position or feed shall be graduated in increments no larger than 0.0001 inch. The feed indicators shall be capable of individually indexing to zero at any point and shall be contiguously calibrated over the full range. Mechanical graduations shall be permanently and legibly engraved on a nonglare background.

3.2.11 Digital readout system. Unless otherwise specified (see 7.2(p)), a digital readout system shall be provided for positional readout of both vertical and horizontal axes. The digital readout system shall consist of an electronic display, related components, and equipment necessary to provide a readily visible digital readout over the full working range. The digital readout shall be capable of indicating in both US and SI systems of measurement. Unless otherwise specified (see 7.2(q)), the digital readout system shall meet the requirements of table V.

TABLE V. Digital readout system requirements.

Parameter	Requirement
Resolution	0.00002 inch
Accuracy	0.0002 inch
Repeatability	0.0001 inch

3.2.12 Lubrication system. A lubrication system shall be provided to ensure lubrication of all moving parts. When specified (see 7.2(r)), the lubrication system shall be completely automatic and shall be provided with interlocks to prevent operation in case of pressure loss.

3.2.13 Coolant system. The coolant system shall provide sufficient coolant flow to prevent workpiece overheating and remove grinding swarf from the work area. The coolant system shall be of sufficient capacity to provide full flow rate during the entire grinding process. Unless otherwise specified (see 7.2(s)), the coolant system shall be of the flood type. The system shall be equipped to allow the operator to direct and control the coolant flow to the work and to flush the base.

3.2.14 Hydraulic system. When a hydraulic system is provided, overpressure protection shall be provided to prevent damage to components. A filter system shall ensure delivery of clean fluid to the hydraulic system. The hydraulic system reservoir shall have a sight gauge to indicate fluid level. The hydraulic fluid temperature shall be maintained within the proper operating range during grinding machine operation. A means shall be provided to drain and clean out the hydraulic reservoir.

3.2.15 Electrical system. The electrical system shall conform to ANSI/National Fire Protection Association (NFPA) ANSI/NFPA-79. The machine shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit; and shall be initially wired for the voltage specified (see 7.2(t)).

3.2.16 Electric motors. The grinding machine electric motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see

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7.2(u)), the motors shall have drip-proof enclosures. Control enclosures shall meet the requirements of NEMA ICS 6.
3.2.17 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(v)).

3.3 Performance. Unless otherwise specified (see 7.2(w)), the grinding machine shall meet the alignment specifications and accuracy requirements shown in tables VI and VII, respectively, when tested in accordance with sections 5.3 through 5.3.4 herein.

TABLE VI. Alignment requirements.

Alignment test performed	Allowable deviation
Worktable or chuck flatness in all directions	0.0001 inch per foot
Axial runout of worktable or chuck	0.0001 inch TIR
Surface of rotary table parallel with its traverse	0.0001 inch per foot
Radial runout of wheel spindle true running	0.0002 inch TIR
Axial runout of wheel spindle	0.0002 inch TIR
Vertical travel of wheelhead square with table	0.004 inch per foot travel

Note: TIR = total indicator readout.

TABLE VII. Performance requirements.

Parameter	Allowable deviation
Parallel surfaces	0.0001 inch per foot
Finished dimensions	0.0001 inch per foot

3.4 Machine dimensions and weight. If required, the machine dimensions (length, width, and height) and machine weight shall meet the specified restrictions (see 7.2(x)).

3.5 Safety and health requirements. The grinding machine shall be designed and manufactured in accordance with ANSI B11.9. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29-CFR 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(y)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(z)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(aa)), the nameplate shall contain the following information:

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a. Nomenclature.
b. Manufacturer's name.
c. Machine model designation.
d. Power input (volts, total amperes, phase, frequency).
e. Short circuit/over-current rating.
f. Contract number or order number.
g. National stock number.
h. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(bb)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user’s health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the grinding machine.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer’s drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(cc)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test conditions. The grinding machine shall be tested in accordance with the environmental conditions recommended by the manufacturer.

5.3.2 Alignment tests. The grinding machine shall be tested for alignment using methods consistent with International Standard Organization (ISO) 230/1. The tests shall be performed both prior to and subsequent to the performance test.

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5.3.3 Operational test. The grinding machine and its auxiliary systems shall be operated in accordance with the manufacturer's standard operating procedures. The machine shall be operated at no load for not less than 2 hours. Proper operation of all controls and features shall be verified during the trial period.

5.3.4 Performance tests. The performance tests shall be conducted in accordance with the manufacturer's standard procedures and recommendations. Four plates shall be placed equidistant apart 1.0 inch from the periphery, and four plates shall be placed equidistant apart 1.0 inch from the center or center hole of the magnetic chuck. The test plates shall be made of steel conforming to American Iron and Steel Institute (AISI) - Society of Automotive Engineers (SAE) 1040 carbon steel. The test shall consist of rough grinding 0.01 inch from the top and bottom surfaces of the test plates, and then finish grinding an additional 0.001 inch from both surfaces. The test results shall be compared to the performance requirements of section 3.3.

5.4 Acceptance. Unless otherwise specified (see 7.2(dd)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. The results of the performance tests shall be compared to the performance criteria required by section 3.3. Failure of the machine to meet any of the performance requirements shall be cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified (7.2(ee)).

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following address:

American National Standards Institute (ANSI).

ANSI B7.1	Safety Requirements for the Use, Care, and Protection of Abrasive Wheels
ANSI B11.9	Safety Requirements for the Construction, Care, and Use of Grinding Machines

Applications for copies should be sent to: American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.

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International Organization for Standardization (ISO)

ISO 230/1 Acceptance Code for Machine Tools

Applications for copies should be sent to: American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

National Fire Protection Association (NFPA)

ANSI/NFPA-79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to: National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

National Electrical Manufacturers Association (NEMA).

NEMA MG-1 Motors and Generators

NEMA ICS-6 Enclosures

Applications for copies should be sent to: National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Roslyn, VA 22209.

Occupational Safety and Health Administration (OSHA)

OSHA 1910 Occupational Safety and Health Standards for General Industry (29 CFR Part 1910)

Applications for copies should be sent to: U. S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type and size of grinding machine (see 2).
c. Minimum workpiece height capacity (see 3.1).
d. Grinding wheel speeds, if different (see 3.2.2).
e. Vertical traverse, if different (see 3.2.2).
f. Grinding wheel(s) required (see 3.2.4).
g. Minimum workpiece weight capacity, if different (see 3.2.5).
h. Tilt, specify degrees, if required (see 3.2.5).
i. Table rotation, if different (see 3.2.5).
j. Magnetic chuck, if different (see 3.2.6).
k. Magnetic pole spacing, if different (see 3.2.6).
l. Wheel dresser, if different (see 3.2.7).
m. Semiautomatic cycle, specify requirements (see 3.2.9.1).

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n. Fully automatic cycle, specify requirements (see 3.2.9.2).
o. Graduations, if different (see 3.2.10.).
p. Digital readout, if different (see 3.2.11).
q. Readout requirements, if different (see 3.2.11).
r. Fully automatic lubrication (see 3.3.12).
s. Coolant system, if different (see 3.2.13).
t. Electrical power requirements, if different (see 3.2.15).
u. Motor enclosure, if different (see 3.2.16).
v. Optional components, specify (see 3.2.17).
w. Alignment and performance, if different (see 3.3).
x. Machine dimensions (length, width, and height) and weight, if different (see 3.4).
y. System of units, if required (see 3.6).
z. Dual system of units, if required (see 3.6).
aa. Nameplate information, if different (see 3.7).
bb. Lubrication information, if different (see 3.8).
cc. Testing requirements, if different (see 5.3).
dd. Acceptance criteria, if different (see 5.4).
ee. Packaging requirements, specify (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

AA59392-XX

- Size: See tables I through III for sizes
- Type: A - Type I, Vertical traversing wheelhead
  B - Type II, Horizontal traversing wheelhead
  C - Type III, Vertical and horizontal traversing wheelhead
- CID number

MILITARY INTERESTS:

Custodians
Air Force - 99
Army - AL
Navy - SH

Reviewers
Air Force - 84
Navy - MC

PREPARING ACTIVITY:
DLA - GS

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